

# Pervasive Physical AI

**Jyotika Athavale**<sup>ID</sup>, IEEE Computer Society

**Dejan Milojcic**<sup>ID</sup>, HPE Labs

**Nita Patel**<sup>ID</sup>, Otis



Digital Object Identifier 10.1109/MC.2025.3630725  
Date of current version: 22 December 2025

0018-9162 © 2025 IEEE. All rights reserved,  
including rights for text and data mining, and training  
of artificial intelligence and similar technologies.

COMPUTER

PUBLISHED BY THE IEEE COMPUTER SOCIETY JANUARY 2026 19



*The year 2025 saw an accelerated artificial intelligence (AI) adoption in every foreseeable and unforeseeable facet of human life and profession. The next year will see increased pervasive adoption of physical AI.*

**F**or our sixth Special Issue of Technology Predictions in October 2025 in *Computer*,<sup>1</sup> we received a large number of high-quality submissions that allowed us to complete two special issues. Those papers that better fit the Outlook model we saved for this issue. Altogether, we received 19 submissions for the special issue and three submissions for the Predictions

column. Each full article had at least three, and many had four reviews. Four papers were rejected, seven papers were accepted for the October issue, and eight for this Outlook issue.

### ABOUT THIS ISSUE

Literally all papers in this issue are artificial intelligence (AI)-related, be it AI-motivated,<sup>A2,A3</sup> AI-driven,<sup>A5,A8</sup> in support of AI,<sup>A1,A4</sup> or supported

by AI.<sup>A6,A7</sup> Compared to the previous year, dominated by large language models (LLMs) and language-based

### DISCLAIMER

The authors are completely responsible for the content in this message. The opinions expressed here are their own.

## APPENDIX: RELATED ARTICLES

- A1. T. Hoefler et al., "In-network collective operations: Game changer or challenge for AI workloads?" *Computer*, vol. 59, no. 1, pp. 24–33, Jan. 2026, doi: [10.1109/MC.2025.3616048](https://doi.org/10.1109/MC.2025.3616048).
- A2. J. Handy and T. M. Coughlin, "New memory architectures will drive future computing," *Computer*, vol. 59, no. 1, pp. 34–42, Jan. 2026, doi: [10.1109/MC.2025.3586208](https://doi.org/10.1109/MC.2025.3586208).
- A3. P. Widener, L. Biven, I. Foster, B. Plale, S. Oral, and R. F. da Silva, "Scientific data management beyond traditional computing boundaries," *Computer*, vol. 59, no. 1, pp. 43–53, Jan. 2026, doi: [10.1109/MC.2025.3589134](https://doi.org/10.1109/MC.2025.3589134).
- A4. K. Post, Z. Yin, M. Olapade, F. Valiyev, P. Nurmi, and H. Flores, "Sensor data copilots: Enabling human-centric pervasive environments," *Computer*, vol. 59, no. 1, pp. 54–63, Jan. 2026, doi: [10.1109/MC.2025.3584589](https://doi.org/10.1109/MC.2025.3584589).
- A5. A. Akintola, J. Ma, N. H. Motlagh, G. Bouloukakis, and H. Flores, "Autonomous indoor customization: The future of human comfort and productivity," *Computer*, vol. 59, no. 1, pp. 64–73, Jan. 2026, doi: [10.1109/MC.2025.3574818](https://doi.org/10.1109/MC.2025.3574818).
- A6. R. Jolak et al., "Navigating the future: Essential considerations for the engineering of software-defined vehicles," *Computer*, vol. 59, no. 1, pp. 74–84, Jan. 2026, doi: [10.1109/MC.2025.3580508](https://doi.org/10.1109/MC.2025.3580508).
- A7. E. B. John and L. K. John, "AI and smart sensors usher in a new era in patient care," *Computer*, vol. 59, no. 1, pp. 85–94, Jan. 2026, doi: [10.1109/MC.2025.3600046](https://doi.org/10.1109/MC.2025.3600046).
- A8. N. U. S. Sabith, S. M. Mamun, M. Rabbani, and S. I. Ahamed, "Toward a noninvasive mHealth platform," *Computer*, vol. 59, no. 1, pp. 95–107, Jan. 2026, doi: [10.1109/MC.2025.3600446](https://doi.org/10.1109/MC.2025.3600446).

AI, this outlook is focused much more on sensors and physical, pervasive AI,<sup>A5,A6,A7,A8</sup> with only one article covering LLMNs.<sup>A4</sup> Not surprisingly, the physical AI is dominated by verticals, such as indoor customization,<sup>A5</sup> software-defined vehicles,<sup>A6</sup> pervasive environments,<sup>A4</sup> and health.<sup>A7,A8</sup>

We have grouped papers into three groups: fundamentals and science, pervasive, and health.

## FUNDAMENTALS AND SCIENCE

In the article “In-Network Collective Operations: Game Changer or Challenge for AI Workloads?” authors Hoefler et al. evaluate in-network collective operations, such as Allgather, Allreduce, Reduce\_scatter, and Alltoall. These operations were developed for high-performance computing and are defined in the message passing interface standard. Authors now explore how these operations can be executed in-network, either at the node or switch, for the AI workloads. The study highlights both the performance benefits and architectural challenges associated with in-network computation for large-scale AI systems.<sup>A1</sup>

The article “New Memory Architectures Will Drive Future Computing,” by Handy and Coughlin, addresses the key dependency of AI: data. AI performance is inherently constrained by the quality and scale of its training data, and with physical AI, the amount of data will explode. The authors discuss the outlook for different types of memory and storage, as well as different interconnects to access this data. They advocate that the key challenge to accessing data, called the memory wall, will be addressed by new data architectures.<sup>A2</sup>

In the article, “Scientific Data Management Beyond Traditional Computing

Boundaries,” Widener et al. address the data wall higher in the stack, offering eight approaches. They advocate fluid data movement, user-centric storage semantics, distributed storage solutions, integration into compute facility infrastructure, active preservation systems, enhanced data protection and control mechanisms, AI data readiness, and data and workflow provenance. Their recommendations provide guidance for funding agencies on structuring next-generation scientific facilities.<sup>A3</sup>

## PERVASIVE

The article “Sensor Data Copilots: Enabling Human-Centric Pervasive Environments,” by Post et al. is the only article in this issue that addresses LLMs and chatbots. These chatbots interact with sensors, converting unstructured data into meaningful insights and thereby better assisting humans in their tasks. They accomplish this based on spatial relationships between devices and users while

offering intuitive interfaces for applications that can leverage this information further.<sup>A4</sup>

The article “Autonomous Indoor Customization: The Future of Human Comfort and Productivity,” Akintola et al. address physical I (robots, sensors) from the human comfort and productivity perspective. They explore indoor spaces, where frequent movement of devices can benefit from autonomous reconfiguration to support such dynamics. Their vision supports proactive, autonomous intelligent devices integrated in the pervasive architecture of indoor spaces.<sup>A5</sup>

In the article “Navigating the Future: Essential Considerations for the Engineering of Software-Defined Vehicles,” Jolak et al. present the evolution of the automotive industry from traditional to software-defined. This enables numerous benefits, such as improved customization and customer experience, increased software-based innovation, connectivity, and sensors utilization for improved performance and safety, and many others.<sup>A6</sup>

## ABOUT THE AUTHORS

**JYOTIKA ATHAVALE** is the functional safety silicon lead at Waymo, Mountain View, CA 94043 USA. Contact her at [jyotika.a.athavale@gmail.com](mailto:jyotika.a.athavale@gmail.com).

**DEJAN MILOJICIC** is an HPE Fellow and a vice president at HPE Labs, Milpitas, CA 95035 USA. His research interests include high-performance computing, operating systems, distributed systems, and systems management. Milojevic received a Ph.D. in distributed systems from the University of Kaiserslautern. He is a Fellow of IEEE. Contact him at [dejan.milojevic@hpe.com](mailto:dejan.milojevic@hpe.com).

**NITA PATEL** is a vice president of engineering at Otis, Farmington, CT 06032 USA. Patel received an MSCpE from Southern Methodist University. Contact her at [nita.patel@ieee.org](mailto:nita.patel@ieee.org).

## GUEST EDITORS' INTRODUCTION

### HEALTH

In the article “Artificial Intelligence and Smart Sensors Usher a New Era in Patient Care,” John and John discuss smart sensors as a basis for health care, based on a unique neural network paradigm called weightless neural networks (WNNs). WNNs can run on tiny devices, such as intelligent band aids. These devices can detect arrhythmia with 88% precision. These devices thereby enable personalized care, early detection, and remote monitoring.<sup>A7</sup>

The article “Toward a Noninvasive mHealth Platform,” by Sabith et al advocates AI-driven health care, which

obviates the need for invasive blood tests, while offering early diagnostics. This technology is based on using mobile phones and leveraging photoplethysmography signals from the fingertip and facial videos. Their work underscores a paradigm shift toward accessible, affordable, and self-managed digital health care.<sup>A8</sup>

**W**e hope you will enjoy this special issue as much as we enjoyed guest editing it. We would appreciate receiving your feedback. It takes a village to prepare a special issue.

### ACKNOWLEDGMENT

We thank the authors and reviewers of all submitted papers. We thank the IEEE Computer Society staff, including Carrie Clark and Abhishek Tiwari. We thank Jeff Voas, the outgoing Editor-in-Chief of *Computer*, for guiding us in every step of the process and for his encouragement. 

### REFERENCES

1. J. Athavale, D. Milojicic, and N. Patel, “Megatrends for the benefit of humanity: AI, health, robotics, and space,” *Computer*, vol. 58, no. 10, pp. 14–18, Oct. 2025, doi: [10.1109/MC.2025.3593393](https://doi.org/10.1109/MC.2025.3593393).



**IEEE COMPUTER SOCIETY**  
**Call for Papers**

Ensure your research is easily discoverable by being indexed in major databases and optimized for search engines.

 **GET PUBLISHED**  
[www.computer.org/cfp](http://www.computer.org/cfp)

 **IEEE**